Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools.

Additional Service Items

- No additional service items required.

Preparatory Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Clean debris from the fittings.
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the electronic control module cooling plate. Refer to Procedure 008-128 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-128.html)

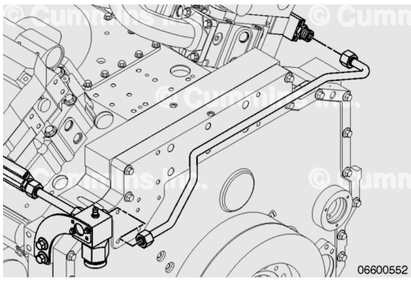
Remove

with Mechanically Actuated Injector

Note : This portion of the Remove section is for engines with mechanically actuated injectors and the new style fuel drain line bracket.

Remove the fuel return line from the engine. Refer to the OEM service manual.

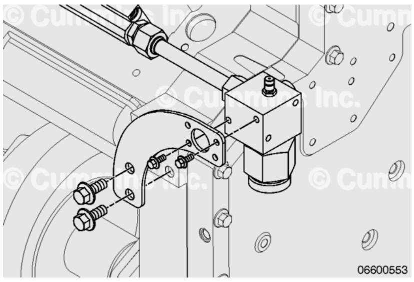
Remove the fuel drain tube.



Note the location and orientation of the check valve and coupling. They **must** be installed in the same location.

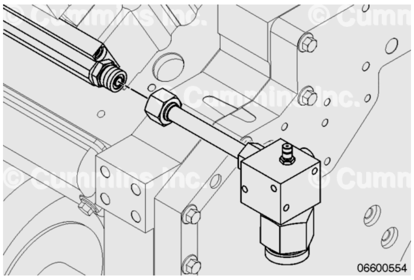
Remove the two capscrews securing the bracket to the tee line connector block.

Remove the two capscrews securing the bracket to the front of the gear housing.

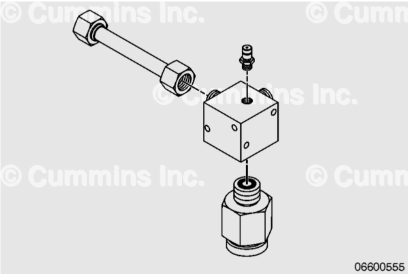


Loosen the fuel drain nut closest to the fuel manifold.

Remove the fuel drain line and tee line connector assembly from the front of the right bank fuel manifold.



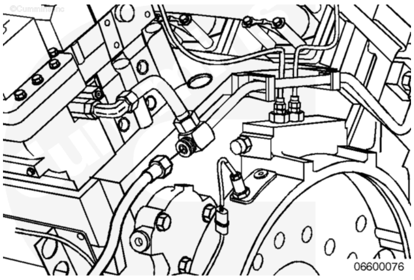
Remove the fuel drain line, check valve, and coupling from the tee line connector. This can require a vise.



Note : This portion of the Remove section is for engines with mechanically actuated injectors and the old style fuel drain line bracket.

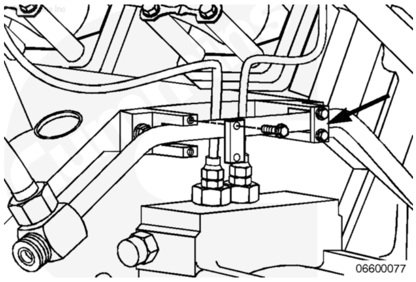
Note : The option illustrated is a left rear fuel drain location. The fuel drain can also be located at the right rear, left front, and right front.

Remove the fuel drain hose from the fuel drain line connector.



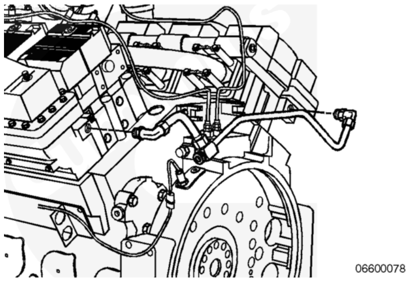
Remove the two capscrews from each fuel drain line support block.

Remove the support block cap from the support block.



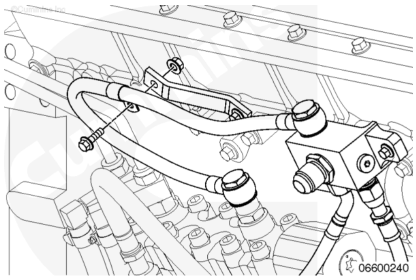
Disconnect the fuel drain line from each fuel manifold.

Remove the fuel drain line assembly.

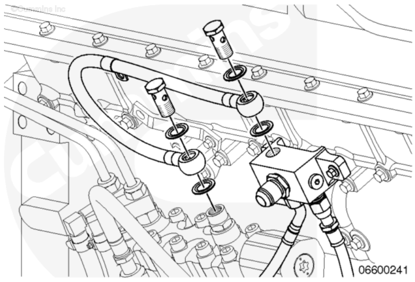


with Electronically Actuated Injector

Remove the capscrew and p-clip which clamp the drain line between the mechanical dump valve and the drain manifold block, to the brace that secures it to the engine.

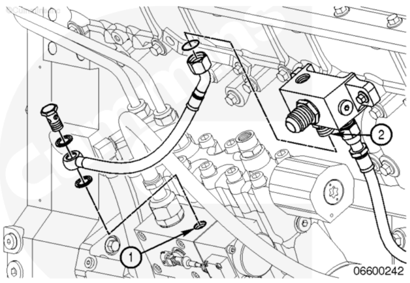


Remove the banjo capscrews and the sealing washers securing the drain line between the mechanical dump valve and the drain manifold block. Discard the sealing washers.

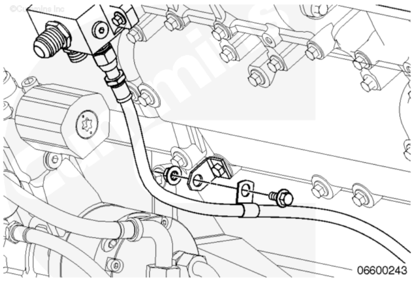


Remove the banjo capscrew and sealing washers securing the air bleed line from the drain manifold block to the fuel pump (1). Discard the sealing washers.

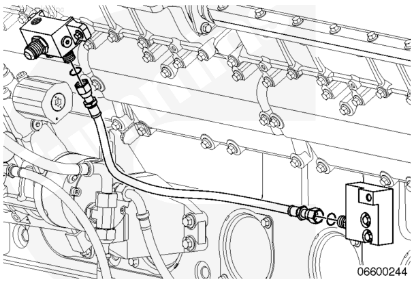
Remove the air bleed line from the drain manifold block (2). Discard the o-ring.



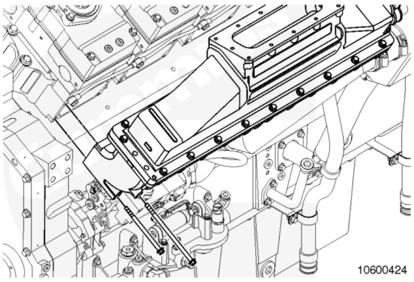
Remove the capscrew, nut, and p-clip which clamp the fuel drain line between the drain manifold block and the fuel block manifold, to the brace that secures it to the engine.



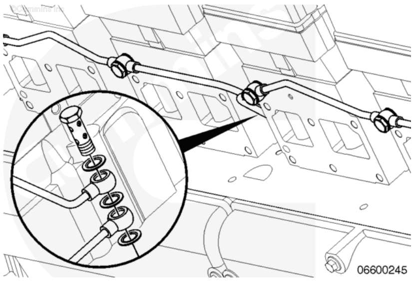
Remove the fuel drain line that connects the drain manifold to the fuel block manifold. Discard the o-rings.



If it is necessary to remove the main fuel manifold, the aftercooler assembly will need to be removed. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)

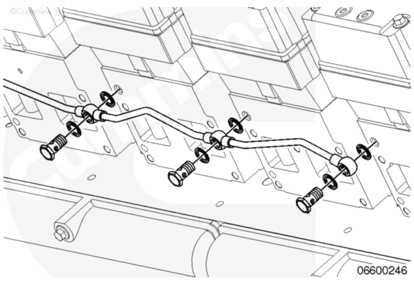


Remove the double banjo capscrew and sealing washers that secure the main fuel drain manifold to the block. Discard the sealing washers.



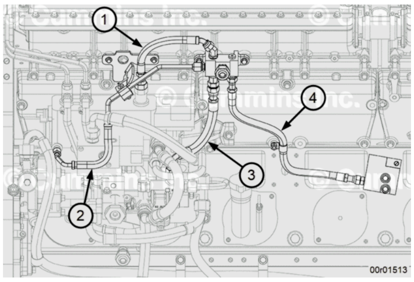
Remove the remaining eight banjo capscrews, sealing washers, and fuel drain manifold. Discard the sealing washers.

Repeat these procedures on the opposite cylinder bank.



For Generator Model DQKAN with Low NOx Emissions Capability

There are four fuel drain lines that connect to the fuel drain manifold:

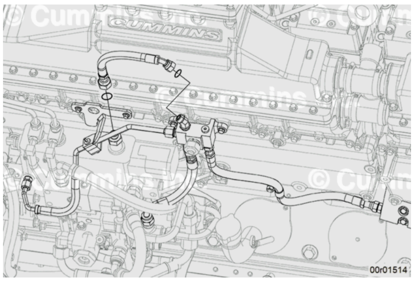


1. Fuel pressure relief valve return
2. High-pressure pump return
3. Fuel manifold Drain line
4. Injector return line

There is an injector return manifold that runs the length of four cylinders

located behind the aftercooler assemblies on each bank.

Disconnect fittings at the fuel pressure relief valve and fuel drain manifold.

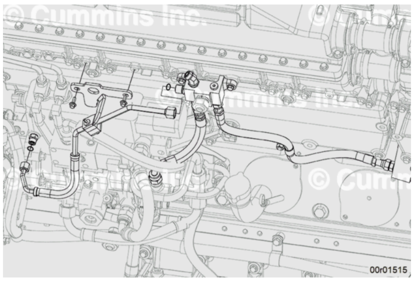


Remove fuel pressure relief valve return line.

Discard O-rings.

⚠ WARNING ⚠

The high-pressure pump return line may be hot. Do not touch the line while the engine is running, and allow adequate time for the line to cool before servicing.



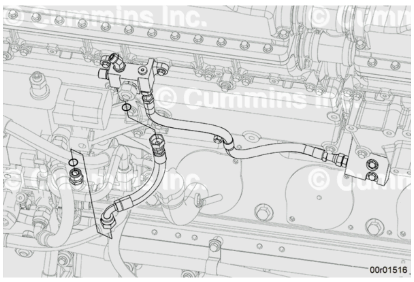
Loosen the block securing the pump return line to the bracket.

Disconnect the fittings securing the pipe to the high-pressure pump and fuel drain manifold.

Discard the o-rings.

Remove the line.

Disconnect the fitting at the fuel pump gerotor and drain manifold block.

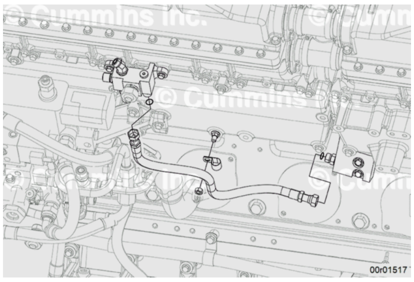


Discard the o-rings.

Remove p-clip from camshaft follower housing.

Disconnect the fitting at the drain manifold block and fuel manifold block.

Discard the o-rings.



The procedure for removal of the main fuel manifold, if necessary, is the same as detailed above for standard product.

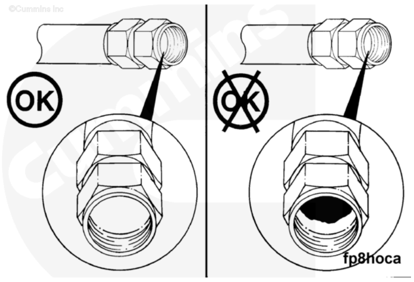
Clean and Inspect for Reuse

with Mechanically Actuated Injector

Inspect the inside of the engine-to-fuel tank hose for the following:

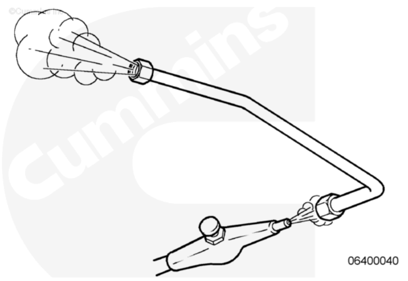
- The inner lining of the hose can separate from the center hose section
- A separation of the inner flap can cause a restriction in the fuel flow.

Inspect for any pinches in the hose that can restrict the flow.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



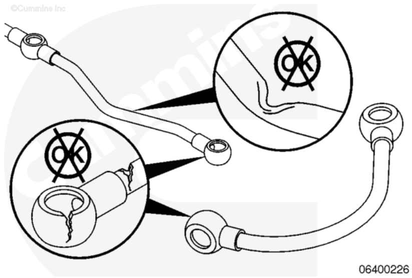
Flush the fuel drain lines and remove any loose dirt particles using compressed air.

Inspect the fuel drain line for cracks or sharp bends that can cause a loss of pressure.

with Electronically Actuated Injector

Inspect the drain lines for cracks and other visible damage. The drain lines **must** be replaced if they are cracked or otherwise damaged.

Inspect the banjo capscrews for visible damage. Replace if damage is found.



Install

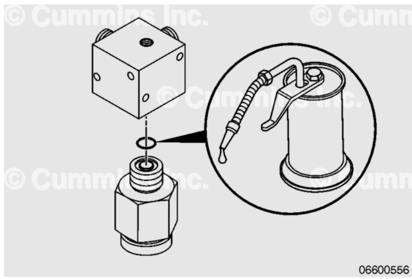
with Mechanically Actuated Injector

Note : This portion of the Install section is for engines with mechanically actuated injectors and the new style fuel drain line bracket.

Check that the o-ring has been fitted to the check valve.

Lubricate the o-ring with clean engine oil.

install the check valve into the tee line connector and tighten. It will be necessary to hold the line connector in a vise.

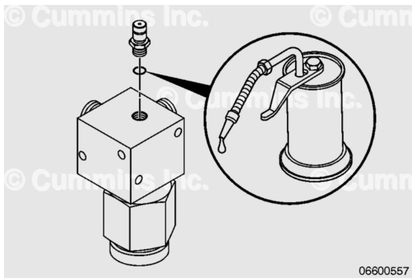


Torque Value: 60 n•m [44 ft-lb]

Install the o-ring to the coupling.

Lubricate the o-ring with clean engine oil.

Install the coupling in the same location and orientation as it was removed.

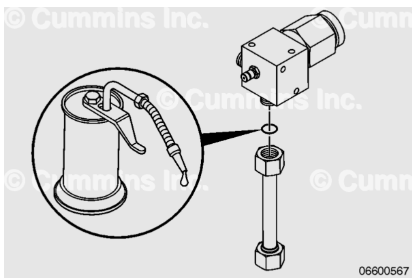


Torque Value:
Fuel Line Coupling
1. 11 n•m [97 in-lb]

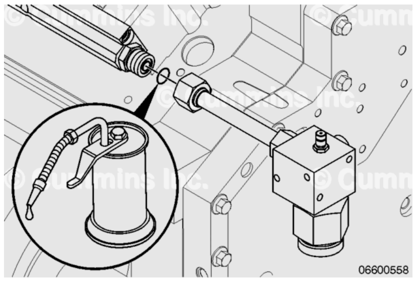
Install an o-ring to the tee line connector where the fuel drain line connects. Use clean engine oil to keep the o-rings in place.

Hold the drain line in the upright position to make sure the o-ring does **not** get dislodged. Install the fuel drain line to the tee line connector.

Hand tighten **only** at this time.



Install an o-ring onto the face of the male connector at the front of the right bank fuel manifold. Lubricate the o-ring with clean engine oil.

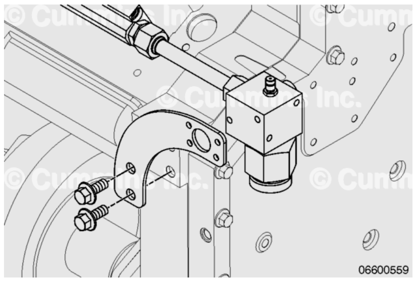


Install the tee line connector assembly to the male connector as shown in the illustration.

Hand tighten **only** at this time.

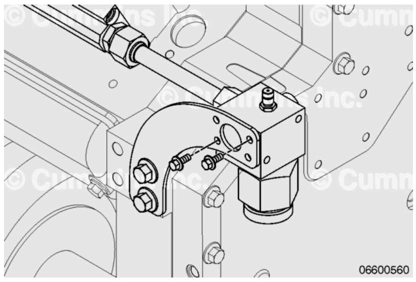
The check valve **must** be in the same location and orientation as it was removed.

Install the support bracket to the front mounting holes in the side of the front gear housing with two mounting capscrews.



Hand tighten **only** at this time.

Install the support bracket to the tee line connector with two mounting capscrews.



Hand tighten **only** at this time.

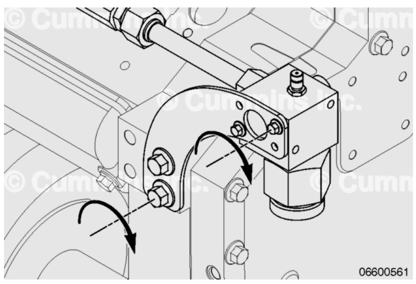
Tighten the fuel drain line nut nearest the fuel manifold, while holding the male connector in place.

Torque Value:
Fuel Drain Line Nut Nearest Fuel Manifold
1. 65 n•m [48 ft-lb]

Tighten the fuel drain line nut nearest the connector block.

Torque Value:
Fuel Drain Line Nut Nearest Connector Block
1. 65 n•m [48 ft-lb]

Tighten the two capscrews that secure the tee line connector to the bracket.



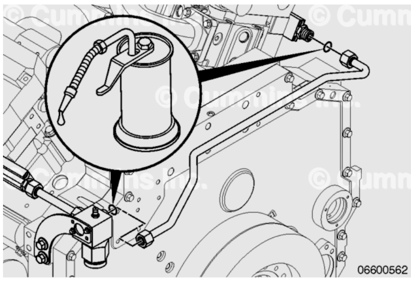
Torque Value:
Bracket to Connector Capscrews
1. 18 n•m [159 in-lb]

Tighten the two capscrews that secure the bracket to the front of the gear housing.

Torque Value:
Bracket to Gear Housing Capscrews

1. 195 n•m [144 ft-lb]

Check the fuel drain line and nuts at each end of the line.



Install an o-ring onto the male connector at the tee line and an o-ring to the male connector at the fuel manifold.

Lubricate the o-rings with clean engine oil.

Install the fuel drain line to the male connector at the tee line. Make sure the o-ring does **not** get dislodged. Tighten the fuel drain line nut at the tee line.

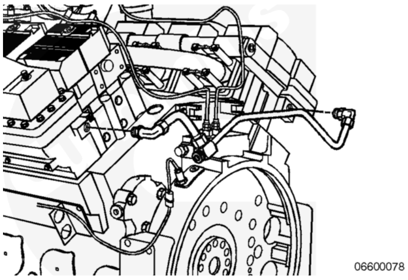
Torque Value: 120 n•m [89 ft-lb]

Install the fuel drain line nut at the fuel manifold male connector and tighten.

Torque Value: 120 n•m [89 ft-lb]

Install the fuel return line. Refer to the OEM service manual.

Note : This portion of the Install section is for engines with mechanically actuated injectors and the old style fuel drain line bracket.

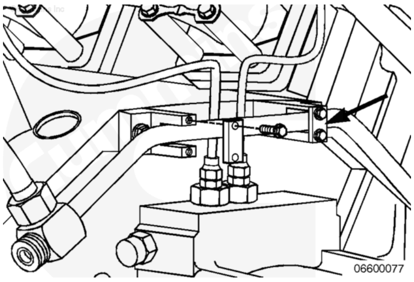


Note : The option illustrated is a left rear fuel drain location. The fuel drain can also be located at the right rear, left front, and right front.

Install and tighten the fuel drain line to the fuel manifold.

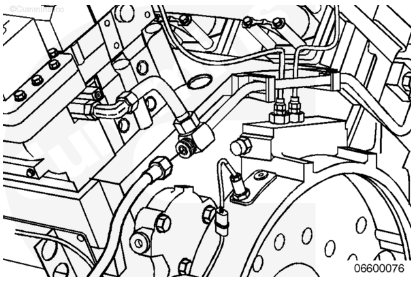
Torque Value: 45 n•m [33 ft-lb]

Install the support block clamp over the fuel drain line and tighten.



Torque Value: 9 n•m [80 in-lb]

Install and tighten the drain hose to the fuel drain line.



Torque Value: 45 n•m [33 ft-lb]

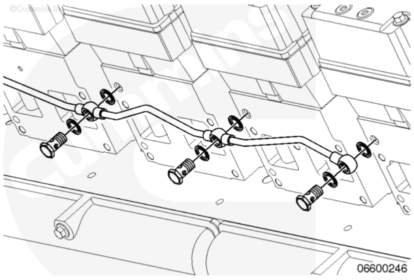
with Electronically Actuated Injector

Install the fuel drain manifold, eight banjo capscrews, and new sealing washers.

Torque Value:

Fuel Drain Manifold Capscrews

1. 30 n•m [22 ft-lb]

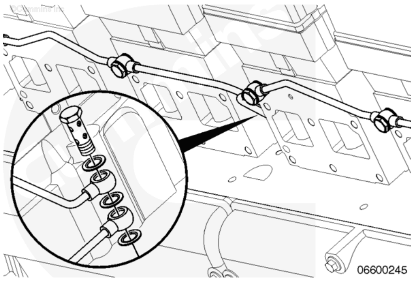


Install the double banjo capscrew and new sealing washers that secure the fuel drain manifold to the block.

Torque Value:

Fuel Drain Manifold

1. 40 n•m [30 ft-lb]



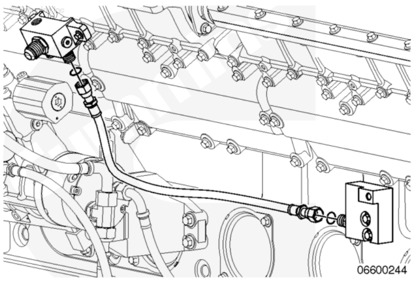
Repeat these procedures on the opposite cylinder bank.

Install the fuel drain line that connects the drain manifold block to the fuel block manifold. Use new o-rings at each end.

Torque Value:

Fuel Drain Line

1. 45 n•m [33 ft-lb]

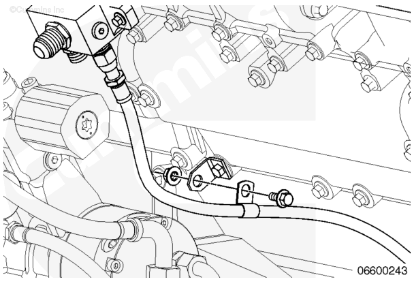


Install the screw, nut and p-clips which clamp the fuel drain line between the drain manifold block and the fuel block manifold, to the brace that secures it to the cylinder block.

Torque Value:

P-Clip Capscrew

1. 22 n•m [195 in-lb]

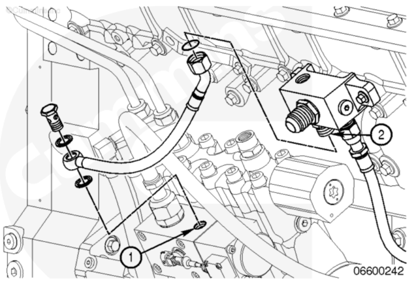


Install the banjo capscrew and new sealing washers securing the air bleed line from the drain manifold block (2) to the fuel pump (1).

Torque Value:

Air Bleed Line

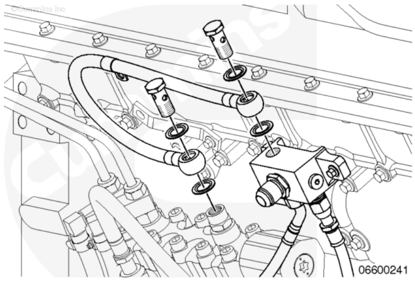
1. 30 n•m [22 ft-lb]



Install the air bleed line to the drain manifold block. Use a new o-ring.

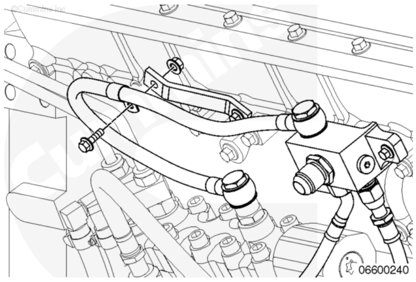
Torque Value:
Air Bleed Line
1. 45 n•m [33 ft-lb]

Install the banjo capscrews and new sealing washers securing the drain line between the mechanical dump valve and the drain manifold block.



Torque Value:
Fuel Drain Line Banjo Capscrew
1. 48 n•m [35 ft-lb]

Install the capscrew and p-clip which clamp the drain line between the mechanical dump valve and the drain manifold block, to the brace that secures it to the cylinder block.



Torque Value:
P-Clip Capscrew
1. 22 n•m [195 in-lb]

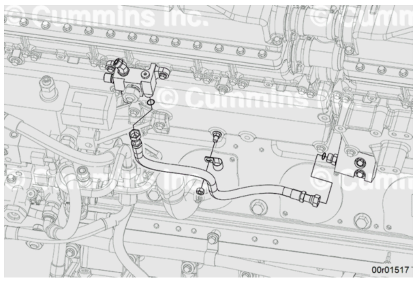
For Generator Model DQKAN with Low NOx Emissions Capability

The procedure for installation of the main fuel manifold, if necessary, is the same as detailed above for standard product.

Install new o-rings.

Connect the fitting at the drain manifold block and fuel manifold block.

Install p-clip to the camshaft follower housing.

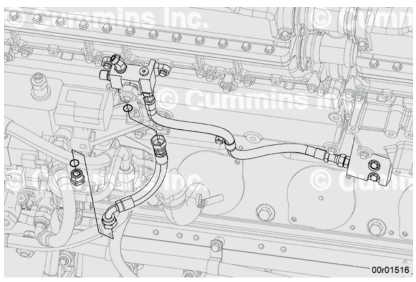


Torque Value:
P-Clip 43 n•m [32 ft-lb]

Torque Value:
Hose Fittings 55 n•m [41 ft-lb]

Install new o-rings.

Install the fitting at the fuel pump gerotor and drain manifold block.



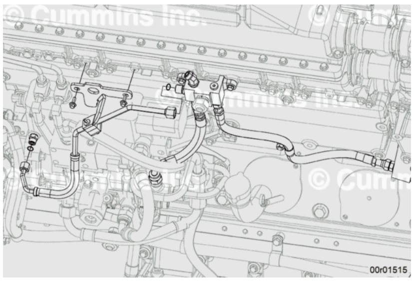
Torque Value:
Hose Fittings 60 n•m [44 ft-lb]

Install new o-rings.

Install the line, securing the fittings at the high-pressure pump and fuel drain manifold hand tight.

Install the line to the block clamp securing the Pump return line to the bracket and tighten.

Torque the fittings securing the line at the high-pressure pump and fuel drain manifold.

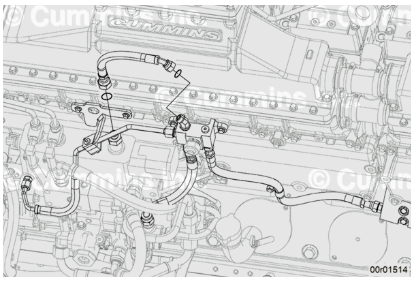


Torque Value:
Hose Fittings 55 n•m [41 ft-lb]

Torque Value:
Hose Bracket 10 n•m [89 in-lb]

Install new o-rings.

Connect and torque the fittings at the fuel pressure relief valve and the fuel drain manifold.



Torque Value:
Hose Fittings 45 n•m [33 ft-lb]

Finishing Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the aftercooler assembly. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Install the Electronic Control Module cooling plate. Refer to Procedure 008-128 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-128.html)

* Reconnect the batteries. See equipment manufacturer service information.
* Operate the engine and check for leaks.

Last Modified: 01-Apr-2021